

Today

Populations vs. Samples (quick)

Statistics and Sample Spaces

Distributions

Quantiles

Distribution Shapes

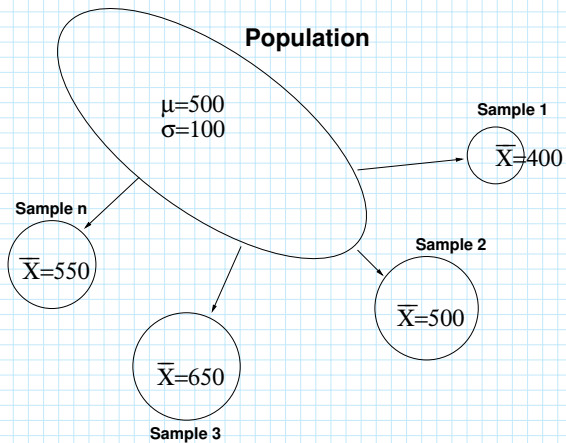
Central Tendency

Populations vs. Samples

- ▶ Populations → Parameters
- ▶ Samples → Statistics

	Statistic	Parameter
Mean	$\bar{X}$	μ
Variance	s^2	σ^2
Correlation	r	ρ

Populations vs. Samples



Descriptive vs. Inferential Statistics

- ▶ Descriptive statistics: describe the characteristics of a group or population
- ▶ Inferential statistics: infer characteristics of a population based characteristics of a sample from that population.

What is a statistic?

A statistic is any number (or numbers) mathematically derived from a data set. It is important that you think of a statistic as a measurement of a dataset, as opposed to the measurements we obtain for an individual in that dataset.

Statistics

Definitions (Sample and Statistic)

A *sample* is a collection of measurements, often written as

$$X = \{X_1, X_2, X_3, \dots, X_N\},$$

where N is the number of measured units (individuals) in the sample. A *statistic* $S(X)$ is any quantity or quantities computed from X :

$$S(X) = f(X_1, X_2, \dots, X_N).$$

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Example

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$$

Variables and Sample Spaces

Sample Space or Support

- ▶ A discrete variable X_i can take on values from a finite or countably infinite set $\{x_1, x_2, \dots\}$.
- ▶ A continuous variable X_i can take on values in a range like $[0, 1]$ or $[0, \infty)$.
- ▶ X_i is a placeholder for individual i 's measurement and x_i is the observed value for that measurement.

Load the Data

```
data <- read.csv("GSS_1980.csv",header=TRUE)
head(data)

# Delete the respondent ID
data$X <- NULL
head(data)

# Give the variables in the dataframe nice names (be careful about order)
names(data) <- c("income","sibs","age","happy","health","taught")

# Compute the frequency and relative frequency distributions
table(data$health)
table(data$health)/length(data$health)
```

Output

```
> data <- read.csv("GSS_1980.csv",header=TRUE)
>
> # Take a peek at the first few lines of the dataset:
> head(data)
      X REALRINC SIBS AGE HAPPY HEALTH YOUNGEN
1 10653      8308   4  62    3      2        1
2 10654     11329  15  77    1      2        2
3 10656     16994   0  28    1      2        2
4 10657     139297  5  24    2      1        1
5 10659      32100  4  63    2      1        1
6 10661      56647  1  62    1      2        2
> # We don't need the respondent ID number so we'll delete it:
> data$X <- NULL
> head(data)
  REALRINC SIBS AGE HAPPY HEALTH YOUNGEN
1      8308   4  62    3      2        1
2     11329  15  77    1      2        2
3     16994   0  28    1      2        2
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5     32100  4  63    2      1        1
6     56647  1  62    1      2        2
> # Give the variables in the dataframe nice names (be careful about order)
> names(data) <- c("income","sibs","age","happy","health","taught")
> # Note that the $ character lets you pull out a variable from a data frame:
> table(data$health) # frequency distribution
 1   2   3   4
319 400 132  22
> table(data$health)/length(data$health) # relative freq distribution
      1      2      3      4
0.36540664 0.45819015 0.15120275 0.02520046
```

GSS "Health" Question

Sample space $S = \{1, 2, 3, 4\}$

	Health Rating			
Raw Frequency	1	2	3	4
$f_X(x)$	319	400	132	22

	Health Rating			
Relative Frequency	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03

GSS "Health" Question

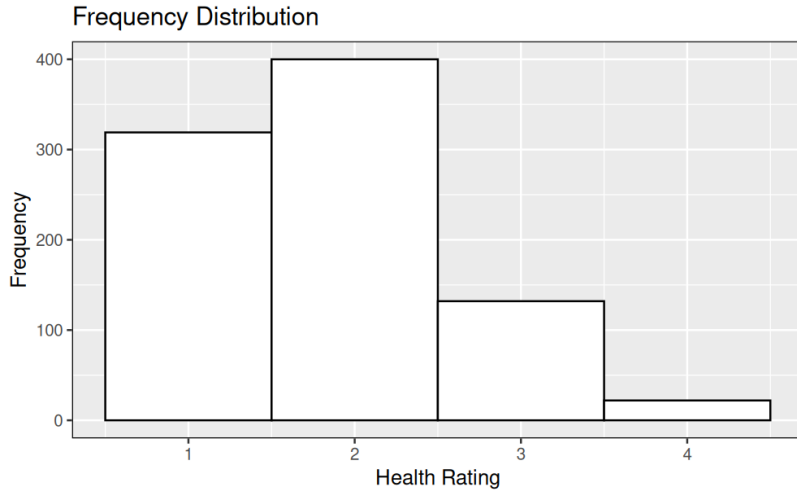
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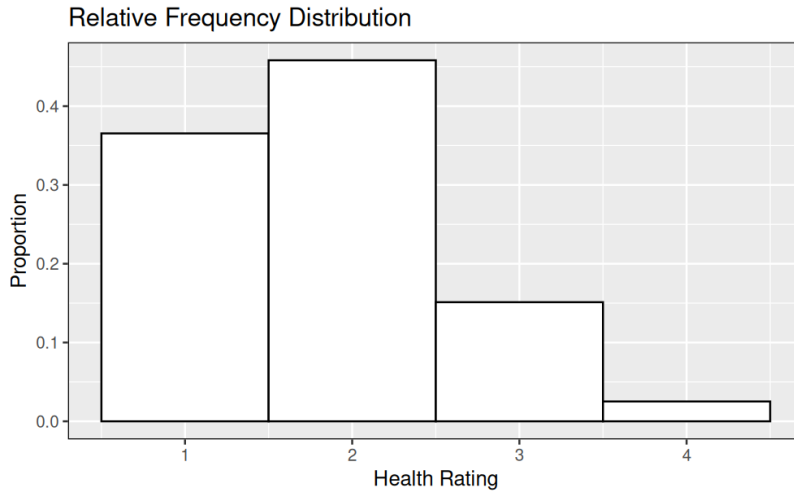
	Health Rating			
Relative Frequency	1	2	3	4
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What is $Pr(X = 1)$?

(Raw) Frequency Distribution



Relative Frequency Distribution



Cumulative Distributions

The number/proportion of observations equal to or below x_i .

$$\text{(Frequency) } F_X(x_i) = \sum_{y \leq x_i} f_X(y)$$

$$\text{(Relative Frequency) } P_X(x_i) = \sum_{y \leq x_i} p_X(y)$$

Cumulative Distributions

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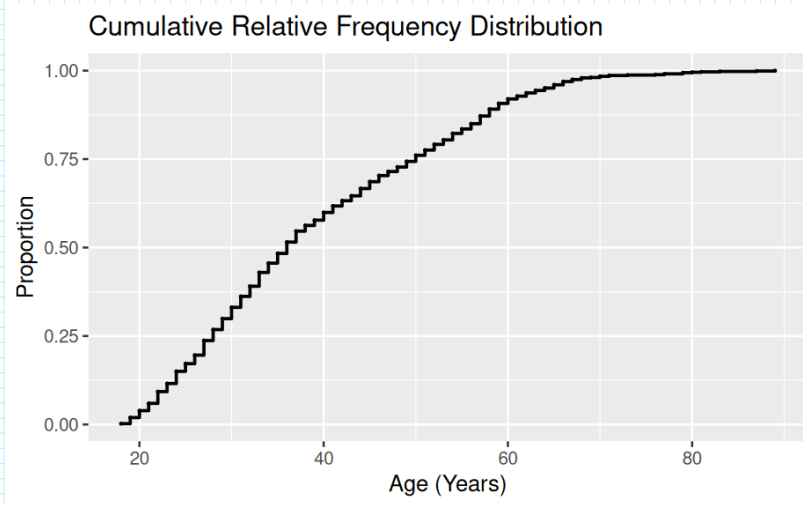
$$\text{(Relative Frequency)} \quad P_X(x_i) = \sum_{y \leq x_i} p_X(y)$$

$$P_X(x) \equiv \Pr(X \leq x)$$

GSS “Health” Question

Relative Frequency	Health Rating			
	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
$P_X(x)$				

Respondent Age



Quantiles

A set of quantiles are the cutpoints that divide a distribution into intervals with equal proportions.

- (1) Median (50%)
- (2) Terciles (33%)
- (3) Quartiles(25%)
- (4) Quintiles (20%)
- (9) Deciles (10%)
- (19) Ventiles (5%)
- (99) Percentiles (1%)

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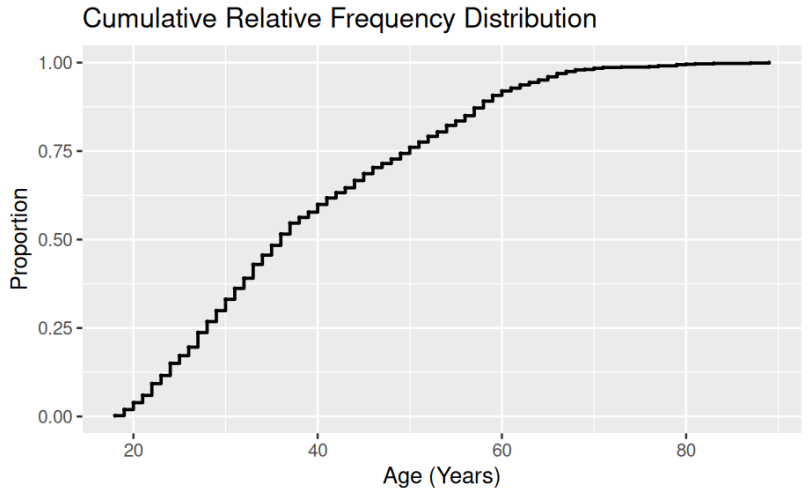
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- (99) Percentiles (1%)

```
# Compute quantiles  
median(data$age)  
quantile(data$age,seq(0,1,by=.05)) # Vary "by=" for other quantiles
```

Respondent Age

Estimate the

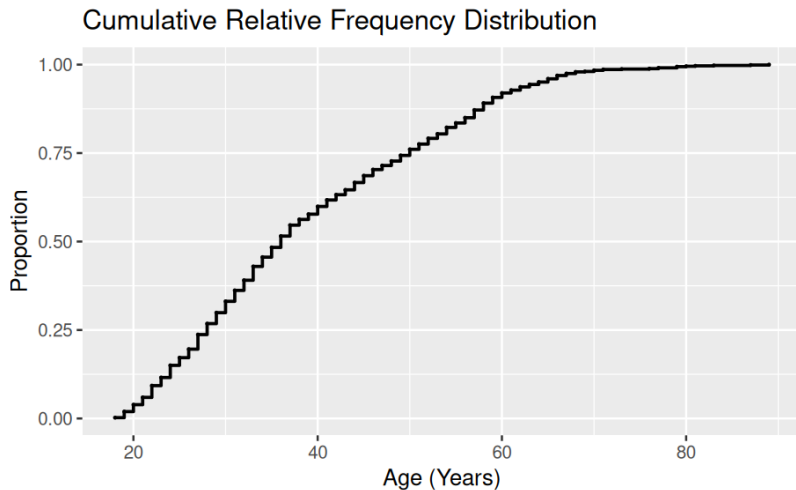
- ▶ median
- ▶ quartiles
- ▶ 45th percentile



Respondent Age

Estimate the

- ▶ median
- ▶ quartiles
- ▶ 45th percentile

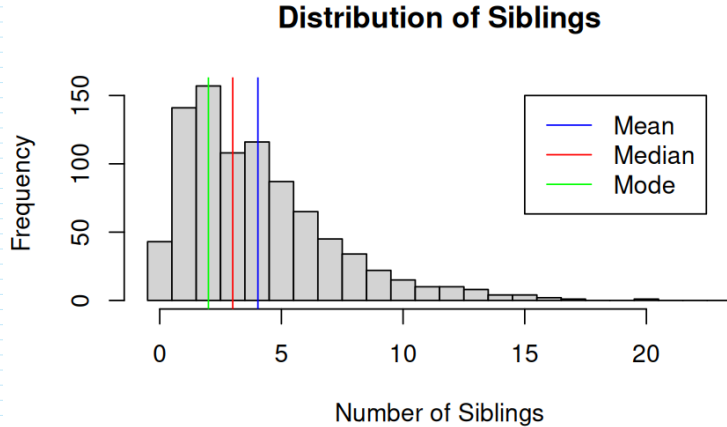


- ▶ What proportion of the respondents are younger than 45?
- ▶ Percentile vs. Percentile Rank

Distribution Shape

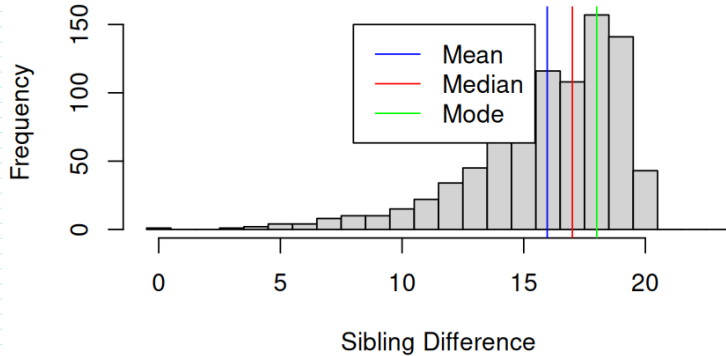
- ▶ Skewness
- ▶ Symmetry
- ▶ Modes

Positively Skewed

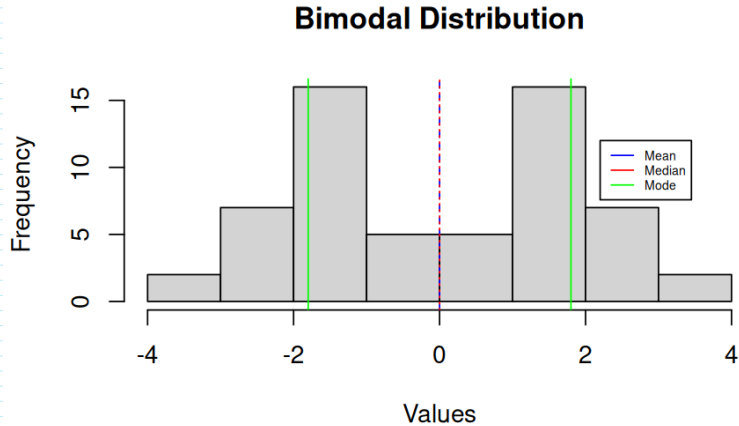


Negatively Skewed

Distribution of Sibling Differences



Symmetric and Bimodal

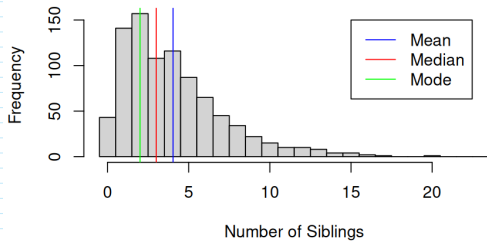


Asymmetric and Bimodal

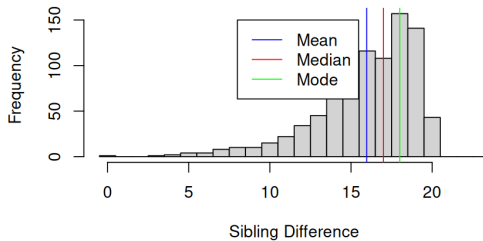
Multimodal

Skewed Distributions

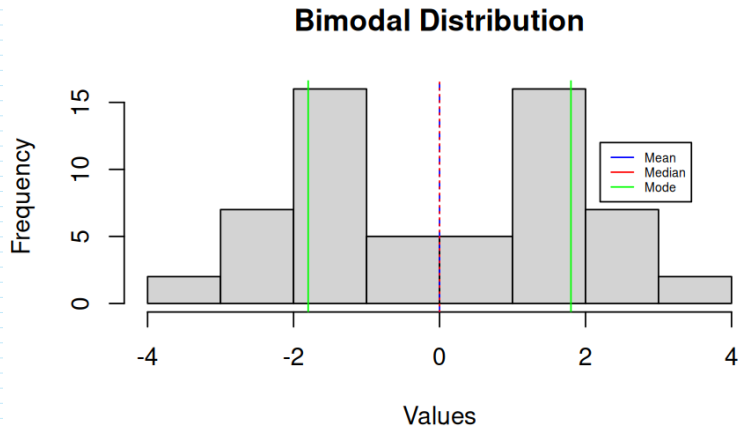
Distribution of Siblings



Distribution of Sibling Differences



Symmetric Distributions



Why Do We Care?

The shape of a distribution determines:

- ▶ what kinds of statistics are most appropriate to describe it.
- ▶ what kinds of statistics are most appropriate for inference.
- ▶ what kinds of statistical *models* we should use for explanation and prediction.

Measures of Central Tendency

The middle or most characteristic score; a measurement of the dataset X .

- ▶ mean:

- ▶ median:

- ▶ mode:

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► mean: $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$.

► median:

► mode:

Measures of Central Tendency

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- ▶ mean: $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$.
- ▶ median: If $Pr(X \leq x) \geq .5$ and $Pr(X \geq x) \geq .5$, then $Md = x$
- ▶ mode:

Measures of Central Tendency

The middle or most characteristic score; a measurement of the dataset X .

- ▶ mean: $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$.
- ▶ median: If $Pr(X \leq x) \geq .5$ and $Pr(X \geq x) \geq .5$, then $Md = x$
- ▶ mode: If $x = \operatorname{argmax}_{x_i} Pr(X = x_i)$, then $Mo = x$.

GSS "Health" Responses

Relative Frequency	Health Rating			
	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
$P_X(x)$	0.37	0.83	0.98	1.00

- ▶ mean
- ▶ median
- ▶ mode

GSS "Health" Responses

Relative Frequency	Health Rating			
	1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
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- ▶ mean
- ▶ median
- ▶ mode

Weighted Means and Expected Values:

$$\bar{X} = \sum_{i=1}^m x_i p_X(x_i)$$

Weighted Means

```
> # Central tendency
> library('DescTools') # Has some useful stuff
> mean(data$health)
[1] 1.836197
> median(data$health)
[1] 2
> Mode(data$health) # Mode function from DescTools package)
[1] 2
attr(,"freq")
[1] 400
>
> # Weighted mean
> p.Health <- table(data$health)
> Health <- as.numeric(names(p.Health))
```

Properties

Shoe Sizes

$$X = \{ \hspace{15cm} \}$$

Properties

Shoe Sizes

Mode

- ▶ Only appropriate measure of central tendency for categorical/nominal data
- ▶ Unstable
- ▶ Not necessarily unique

Properties

Shoe Sizes

Median

- ▶ Appropriate for ordinal or higher scales of measurement
- ▶ Robust to outliers
- ▶ Not necessarily unique
- ▶ Minimizes the Mean Absolute Deviation (MAD)

Properties

Shoe Sizes

Mean

- ▶ Appropriate for interval or higher scales of measurement
- ▶ Very sensitive to outliers

Sensitivity to Outliers

Robert Pershing Wadlow (1918-1940)



Deviation Scores

$$D_i = X_i - \bar{X}$$

- ▶ What is $\sum_{i=1}^N D_i$?
- ▶ What is $\sum_{i=1}^N D_i^2$?

Sum of Squares

$$\sum_{i=1}^N D_i^2 = \sum_{i=1}^N (X_i - \bar{X})^2$$